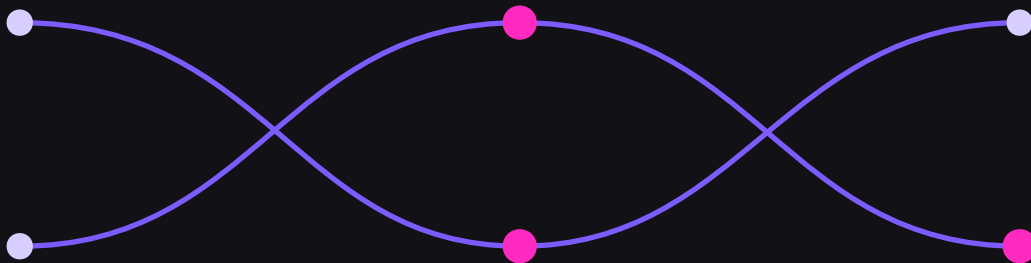


When Machines Start Borrowing the Logic of Life

The emerging convergence of Physical AI, bionics and biology

A designed article on robotics, embodiment and the changing boundary between engineered and living systems.



The emerging convergence of Physical AI, bionics and biology

For decades, the body was the boring part of a robot.

You chose the sensors, bolted on the actuators, and spent your ambition on the software. Intelligence was something you built. The body was something you bought.

That order of operations is starting to break.

In my previous article on Physical AI, I argued that the field is much bigger than the humanoids dominating the conversation. This series follows one thread of it: what happens when the body itself starts borrowing from biology.

A robot swims using living muscle. An arm borrows the octopus's distributed sensing. A prosthetic knee anchors to bone and wires into the user's own muscles. Algorithms help design creatures made entirely of living cells.

These come from different fields: soft robotics, neuroprosthetics, biohybrid robotics, developmental biology. Increasingly, they are the same conversation.

I call it Bionics Physical AI. It is my working label, not an established discipline: physical systems where biological principles, interfaces or living components contribute directly to sensing, actuation, control or embodiment. The label is mine. The research behind it is very real.

Physical AI has a body problem

Most talk about Physical AI starts with the brain. Better foundation models, better perception, better spatial reasoning, better reinforcement learning.

Fair enough. The AI boom was computational, and robotics inherited its vocabulary, including the classic loop of sense, compute, decide, act.

Biology doesn't work like that. An animal's behaviour comes from nervous system, muscles, tendons, skin, joints and environment working together. Shape matters. Elasticity matters. Friction does a surprising amount of the thinking.

That is the idea behind morphological computation and physical intelligence: useful capability can be spread across software, morphology and materials instead of sitting in one central controller.^[1] Researchers have even used a soft robot's own body dynamics as a computational resource.^[2]

Take that seriously and the design question flips. The body decides how hard the control problem is before the software ever sees it.

Borrowing more than shapes

Engineers have always copied nature's shapes. Birds gave us flight, hands gave us grippers, animals gave us walking machines.

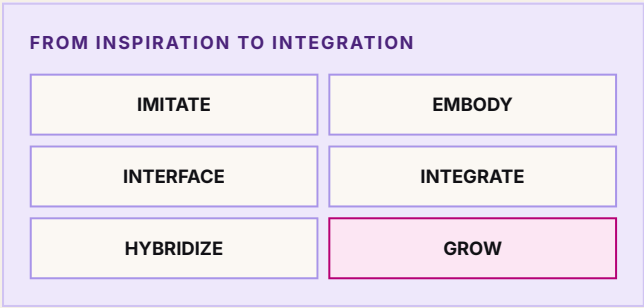
The newer work copies how nature is organised.

In 2026, researchers reported an octopus-inspired soft arm with ten sensorized suction cups. It did not route every touch through a single central processor. Some contact could be handled locally, with reflex-like responses, before higher-level control got involved.^[3]

Not a robot that looks like an octopus arm. A robot that distributes its thinking like one.

From inspiration to integration

Across the literature, biology and robotics meet at several layers:



At one end, engineers copy biological movement, structure or sensing. Further along, biological principles are built into materials and distributed control. Machines connect directly to muscles, nerves and brains. Living muscle, neurons or skin become working parts of a machine. Existing organisms are fitted with electronics. And at the far edge, the physical system is built from living cells.

The layers overlap and develop in parallel. They are useful because they make one direction visible: biology is moving closer to the centre of the engineering process. Recent reviews already describe systems that combine living muscle, neurons, biomaterials and microelectronics, with "intelligent biomachines" emerging as a research direction.^[4]

When the machine becomes part of you

Prosthetics show this convergence most clearly.

In 2025, researchers reported a tissue-integrated bionic knee: direct skeletal attachment, surgically reconstructed muscle interfaces and implanted electronics. Signals from the residual muscles controlled the powered joint.^[5] The knee is still an artificial device. But it is anchored into the skeleton and sits in a control loop that starts in the user's own neuromuscular system.

The same year, an AI "copilot" shared control with a person with paralysis through a non-invasive brain-computer interface. Together they completed a robotic pick-and-place task that the interface alone could not.^[6]

That is a long way from a person simply operating a device. Article 4 goes deeper.

And then the machine gets muscles

Now follow the convergence in the other direction: living tissue inside machines.

In 2025, a team demonstrated an 18-centimetre robotic hand driven by bundles of cultured human skeletal muscle.^[7] In 2026, a thin swimming robot was powered by engineered skeletal-muscle films on a hydrogel body.^[8]

Living tissue changes the engineering problem at once. Muscle has to stay viable. It has to mature. Its performance drifts over time, and it has to connect to synthetic structures without being damaged by them. That is one reason soft electronics is becoming central to biohybrid robotics: flexible electrodes and sensors can move and deform with the tissue.^[9]

A robot built this way needs more than a mechanical architecture. It needs something like a physiology. Article 3 is about exactly that.

The animal becomes the platform

Cyborg-insect research turns the architecture around again.

In 2025, researchers coordinated Madagascar hissing cockroaches fitted with miniature electronic controllers, steering a group through obstructed terrain.^[10] The appeal is easy to see. Evolution already supplied the locomotion, balance, sensing and efficiency. Engineers only add control.

It also brings an ethical question conventional robotics rarely faces so directly. The platform is a living animal, and the capabilities being exploited belong to an organism with its own existence.

What if we grow the physical system?

Xenobots push the idea furthest. In 2020, researchers used evolutionary computation to search for useful body designs, then built selected ones from frog cells. They could move and show simple collective behaviour.^[11] The AI worked upstream, searching for form. The cells did the rest.

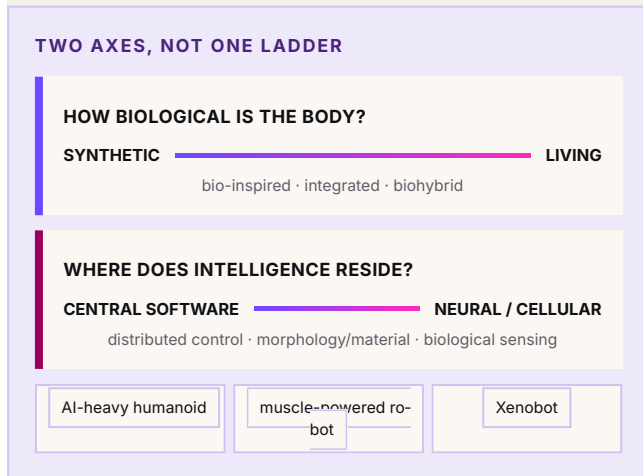
Then came Anthrobots: adult human airway cells that self-organised into motile multicellular structures, without genetic modification or a manually imposed scaffold.^[12]

The final body is no longer specified cell by cell. Researchers set conditions, and the cells help build the structure themselves. Design, fabricate and assemble starts to look more like cultivation. Article 3 picks that up.

Two axes, not one ladder

A muscle-powered robot can be biologically sophisticated with hardly any artificial intelligence. An Anthrobot sits far toward the biological end with no conventional AI controller at all. A humanoid running a powerful multimodal model can have enormous computational capability and remain almost entirely synthetic.

So one ladder isn't enough. I find two dimensions more useful.



A Xenobot sits in an odd place: AI contributes to the design, while the deployed system is cellular. A future biohybrid machine could score high on both axes. That is where this gets interesting.

The boundary is the story

Robotics is borrowing from developmental biology. Materials science is borrowing from physiology. Prosthetics is moving closer to neural engineering. Electronics is getting softer, because tissue is soft. And biology is increasingly something engineers design with.

The first generation of Physical AI gave intelligence a body. Bionics asks what that body should be made of, how it should sense and adapt, and how much of its capability should come from biology itself.

Most of the field will stay rigid, manufactured and synthetic for a long time. And the edges are already blurring. Both can be true.

That is where this series starts.

Which are you picturing when you think about the next generation of robots: a better brain, or a different kind of body?

Next: how much intelligence the body can carry by itself.

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